Equal Temperament Pitch & Frequency Conversions

Mathematical Models of `[mtof]` and `[ftom]` Nodes

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Abstract

This document formulates the logarithmic and exponential relationships between MIDI Note numbers $m \in [0, 127]$, fundamental frequencies $f \in \mathbb{R}^+$ (in Hz), microtonal cent tuning ratios, and Doppler frequency shifts under relativistic coordinate time dilation.

1 MIDI Note to Frequency Conversion (`[mtof]`)

In 12-tone equal temperament (12-TET) with concert pitch reference $A_4 = 440\,\text{Hz}$ at MIDI Note $m = 69$, the fundamental frequency $f(m)$ is defined exponentially:

$$f(m) = 440 \cdot 2^{\frac{m-69}{12}} \tag{1}$$

For an arbitrary reference concert pitch $A_4 = f_0$:

$$f(m, f_0) = f_0 \cdot 2^{\frac{m-69}{12}} \tag{2}$$

2 Frequency to MIDI Note Conversion (`[ftom]`)

The inverse mapping from fundamental frequency $f > 0$ to MIDI Note number $m(f)$ is derived by taking the base-2 logarithm:

$$m(f) = 69 + 12 \cdot \log_2 \left(\frac{f}{440} \right) \tag{3}$$

Using standard natural logarithms:

$$m(f) = 69 + \frac{12}{\ln 2} \cdot \ln \left(\frac{f}{440} \right) \tag{4}$$

3 Cent Tuning & Transposition Ratios

The frequency multiplier ratio $R(c)$ for a microtonal interval of c cents ($100\,\text{cents} = 1\,\text{semitone}$) is:

$$R(c) = 2^{\frac{c}{1200}} \tag{5}$$

Given a base frequency f_{base} and semitone shift Δm with fine detune cents c, the modulated frequency is:

$$f_{\text{final}} = f_{\text{base}} \cdot 2^{\frac{\Delta m + c/100}{12}} \tag{6}$$

4 Relativistic Doppler Pitch Shift under Time Warping

When an audio node is driven by a time dilation factor $\gamma(t)$, the apparent perceived playback frequency $f_{\text{perceived}}$ experiences an acoustic Doppler shift:

$$f_{\text{perceived}}(t) = f_{\text{base}} \cdot |\gamma(t)| \tag{7}$$

Under retrograde time flow ($\gamma(t) < 0$), the frequency magnitude remains $|f_{\text{perceived}}| = f_{\text{base}} \cdot |\gamma(t)|$ while the phase trajectories invert direction.